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Soal 1:

DM::OJ PROBLEMS SUBMISSIONS USERS CONTESTS ABOUT

CCC '07 J4 - Anagram Checker

Canadian Computing Competition: 2007 Stage 1, Junior #4

An anagram is a word or a phrase formed by rearranging the letters of another phrase such as `ITEM` and `TIME`. Anagrams may be several words long such as `CS AT WATERLOO` and `COOL AS WET ART`. Note that two phrases may be anagrams of each other even if each phrase has a different number of words (as in the previous example). Write a program to determine if two phrases are anagrams of each other.

Input Specification

The program should take two phrases, each on a separate line. You may assume that the input only contains uppercase letters and spaces.

Output Specification

The program will print out one of two statements: `Is an anagram.` or `Is not an anagram.`

Sample Input

```
CS AT WATERLOO
COOL AS WET ART
```

Sample Output

```
Is an anagram.
```

Sumber: [CCC '07 J4 - Anagram Checker - DMOJ: Modern Online Judge]

Source code:

```python
kal1 = input("Enter first word: ")
kal2 = input("Enter second word: ")

kal1 = "".join(kal1.split())
urut1 = sorted(kal1)

kal2 = "".join(kal2.split())
urut2 = sorted(kal2)

if urut1 == urut2:
    print("Is an anagram.")
else:
    print("Is not an anagram.")
```

Result:

```
PS C:\Users\Acer\OneDrive\Dokumen\belajar-github> C:\Users\Acer\AppData\Local\Programs\Python\Python313\python.exe c:/Users/Acer/OneDrive/Dokumen/belajar-github/bonus/soal1.py
Enter first word: CS AT WATERLOO
Enter second word: COOL AS WET ART
Is an anagram.
PS C:\Users\Acer\OneDrive\Dokumen\belajar-github> C:\Users\Acer\AppData\Local\Programs\Python\Python313\python.exe c:/Users/Acer/OneDrive/Dokumen/belajar-github/bonus/soal1.py
Enter first word: PAK KAREL
Enter second word: REAL KAPP
Is not an anagram.
```

Soal 2:

CCC '17 S2 - High Tide, Low Tide

Canadian Computing Competition: 2017 Stage 1, Senior #2

Joe Coder is camping near the Bay of Fundy between Nova Scotia and New Brunswick. When he arrived at the bay, he was told that the difference in height between high tide and low tide at the Bay of Fundy was the largest tidal difference in the world. Ever the skeptic, Joe decided to verify this. He chose a reference point and, after learning from the radio when the tides were highest and lowest, he went with a boat to his reference point and measured the depth of the water. Unfortunately, on the last day of his trip, a strong wind scattered his measurements.

Joe has recovered all of his measurements, but they may not be in their original order. Luckily, he remembers some things about his measurements:

- He started measuring water levels at a low tide, his second measurement was of the water level at high tide, and after that the measurements continued to alternate between low and high tides.
- All high tide measurements were higher than all low tide measurements.
- Joe noticed that as time passed, the high tides only became higher and the low tides only became lower.

Given Joe's measurements in no particular order, you must reconstruct the correct order in which the measurements were taken.

Input Specification

The first line contains the integer N $(1 \leq N \leq 100)$. The next line contains N distinct space-separated positive integers, where each integer is at most $1\,000\,000$.

Output Specification

Output the N integers in the unique order that Joe originally took the measurements.

Sample Input

```
8
10 50 40 7 3 110 90 2
```

Sample Output

```
10 40 7 50 3 90 2 110
```

Explanation for Sample Output

The low tide measurements (in order) were 10, 7, 3, and 2.
The high tide measurements (in order) were 40, 50, 90, and 110.

Sumber: [CCC '17 S2 - High Tide, Low Tide - DMOJ: Modern Online Judge]

Source code:

```python
jumlah_angka = int(input("Masukkan jumlah angka: "))
angka = input("Masukkan angka: ")

simpan_angka = []
kecil = []
besar = []
hasil = ""

angka = angka.split()
for i in angka:
    simpan_angka.append(int(i))

simpan_angka = sorted(simpan_angka)

if jumlah_angka % 2 == 0:
    temp = int(jumlah_angka/2)
    for i in range(temp):
        kecil.append(simpan_angka[i])
    kecil = sorted(kecil)[::-1]
    for j in range(temp, jumlah_angka):
        besar.append(simpan_angka[j])
    for k in range(temp):
        hasil += f"{kecil[k]} {besar[k]} "
else:
    temp = int((jumlah_angka-1)/2)
    for i in range(temp+1):
        kecil.append(simpan_angka[i])
    kecil = sorted(kecil)[::-1]
    for j in range(temp+1, jumlah_angka):
        besar.append(simpan_angka[j])
    for k in range(temp):
        hasil += f"{kecil[k]} {besar[k]} "
    hasil += f"{kecil[-1]}"

print(hasil)
```

Result:

```
PS C:\Users\Acer\OneDrive\Dokumen\belajar-github> & C:\Users\Acer\AppData\Local\Programs\Python\Python313\python.exe c:/Users/Acer/OneDrive/Dokumen/belajar-github/bonus/soal2.py
Masukkan jumlah angka: 5
Masukkan angka: 12 53 73 234 14
53 73 14 234 12
PS C:\Users\Acer\OneDrive\Dokumen\belajar-github> & C:\Users\Acer\AppData\Local\Programs\Python\Python313\python.exe c:/Users/Acer/OneDrive/Dokumen/belajar-github/bonus/soal2.py
Masukkan jumlah angka: 8
Masukkan angka: 10 50 40 7 3 110 90 2
10 40 7 50 3 90 2 110
```

Soal 3:

CCC '12 S2 - Aromatic Numbers

Canadian Computing Competition: 2012 Stage 1, Senior #2

This question involves calculating the value of *aromatic* numbers which are a combination of Arabic digits and Roman numerals.

An aromatic number is of the form $ARARAR\ldots AR$, where each A is an Arabic digit, and each R is a Roman numeral. Each pair AR contributes a value described below, and by adding or subtracting these values together we get the value of the entire aromatic number.

An Arabic digit A can be 0, 1, 2, 3, 4, 5, 6, 7, 8 or 9. A Roman numeral R is one of the seven letters I, V, X, L, C, D, or M. Each Roman numeral has a base value:

Symbol	I	V	X	L	C	D	M
Base value	1	5	10	50	100	500	1000

The value of a pair AR is A times the base value of R. Normally, you add up the values of the pairs to get the overall value. However, wherever there are consecutive symbols $ARA'R'$ with R' having a *strictly bigger* base value than R, the value of pair AR must be *subtracted* from the total, instead of being *added*.

For example, the number $3M1D2C$ has the value $3 \times 1000 + 1 \times 500 + 2 \times 100 = 3700$ and $3X2I4X$ has the value $3 \times 10 - 2 \times 1 + 4 \times 10 = 68$.

Write a program that computes the values of aromatic numbers.

Input Specification

The input is a valid aromatic number consisting of between 2 and 20 symbols.

Output Specification

The output is the decimal value of the given aromatic number.

Sample Input 1

```
3M1D2C
```

Output for Sample Input 1

```
3700
```

Sample Input 2

```
2I3I2X9V1X
```

Output for Sample Input 2

```
-16
```

Sumber: https://dmoj.ca/problem/ccc12s2

Source code:

```python
kombo = input("Input: ")
temp =[]
hasil = 0

for i in range(len(kombo)):
    if kombo[i] == "I":
        temp.append("1")
    elif kombo[i] == "V":
        temp.append("5")
    elif kombo[i] == "X":
        temp.append("10")
    elif kombo[i] == "L":
        temp.append("50")
    elif kombo[i] == "C":
        temp.append("100")
    elif kombo[i] == "D":
        temp.append("500")
    elif kombo[i] == "M":
        temp.append("1000")
    else:
        temp.append(kombo[i])

for j in range(len(temp)):
    temp[j] = int(temp[j])

for k in range(1, len(temp)-1, 2):
    if temp[k] >= temp[k+2]:
        hasil += temp[k] * temp[k-1]
    elif temp[k] < temp[k+2]:
        hasil -= temp[k] * temp[k-1]
hasil += temp[-1] * temp[-2]

print(hasil)
```

Result:

```
PS C:\Users\Acer\OneDrive\Dokumen\belajar-github> & C:\Users\Acer\AppData\Local\Programs\Python\Python313\python.exe c:/Users/Acer/OneDrive/Dokumen/belajar-git
hub/bonus/soal3.py
Input: 3M1D2C
3700
PS C:\Users\Acer\OneDrive\Dokumen\belajar-github> & C:\Users\Acer\AppData\Local\Programs\Python\Python313\python.exe c:/Users/Acer/OneDrive/Dokumen/belajar-git
hub/bonus/soal3.py
Input: 2I3I2X9V1X
-16
```